

The Embedding Confound in Linear-Probe Interpretability

Anonymous ACL submission

Abstract

- **Topic.** Linear-probe interpretability identifies low-dimensional residual-stream directions for truth, refusal, sentiment, and sycophancy in chat LLMs, with evidence resting on either probe correlation or directional ablation.
- **Motivation.** A read-layer probe correlation can be substantially inherited from the input embedding’s projection along the same direction — a confound for which no standard held-out control exists.
- **Main contribution.** We audit four published single-direction claims across seven instruction-tuned models with an embedding-partialled correlation control (Eq. 3), a random-direction null, and held-out ablation, and find that the Ardit refusal direction loses 63–66% of its predictive correlation on Llama-3-8B-Instruct and Phi-3-mini under the control — yet directional ablation still suppresses refusals on the same two cells ($\Delta y = -0.48$ and -8.82 , both significant). Linear and causal redundancy come apart.
- **Technical detail.** For each (direction, model) cell we fit a mean-difference direction at a read layer and at the mean-pooled embedding layer L0 on a train fold, then report raw r , L0 r , and partial r on a held-out test fold with 2000-resample bootstrap CIs (Eqs. 1–3). Directional ablation projects the direction out at every layer (Eq. 6); behavioral outcomes are refusal log-odds (Eq. 4) and truth-elicitation score (Eq. 5).
- **Evidence / secondary.** Truth passes 3/3 (drop $\leq 1\%$), sentiment passes 6/7 (drop 3–14%), and sycophancy is degenerate 6/6 under fixed-layer mean-difference. The refusal FAIL survives a random-direction null (observed lr_{partial} at 64.5th and 85th percentiles), 20/20 disjoint-split refits, Holm-Bonferroni correction, and two behavioral metrics.

- **Limitation.** The refusal FAIL is dataset- and checkpoint-conditioned: re-fitting on CAA-style prompts restores PASS on Llama-3-8B (partial $r = +0.85$), and a held-out-family replication (OLMo-2, Yi-1.5, Falcon-3, Tulu-3) returns 0/4 FAIL. The sycophancy result is a predictive-correlation negative, not a causal falsification of CAA.
- **Narrow impact.** Single-direction claims should report raw r , L0 r , and partial r with bootstrap intervals as standard practice. Probe correlation alone is not load-bearing evidence that a direction is causal.
- **Broad impact.** A cheap control (one forward pass at L0 + one direction fit + one partial correlation) can catch a confound that otherwise passes through standard validity checks, strengthening the evidence standard for linear-probe interpretability across the field.

1 Introduction and Related Work

- Open with the recurring interpretability move: identify a low-d residual direction, name it “truth” (Marks and Tegmark, 2023), “refusal” (Arditi et al., 2024), “sentiment” (Tigges et al., 2023), or “sycophancy” (Panickssery et al., 2024).
- Two evidence types dominate: linear-probe correlation OR directional-ablation. Neither is held out against an *input-embedding* confound.
- The gap (one line, McEnerney problem statement): a read-layer probe correlation can be inherited from the input embedding’s projection along the same axis. No standard audit catches this.
- Why the gap matters: published correlations are read as evidence the model “builds” the representation through depth. In two of our

086	cells, that reading is wrong even though the	(2 FAIL / 2 PASS main + 4/4 held-out PASS),	133
087	direction is causally effective.	sycophancy DEGENERATE 6/6 under fixed-	134
088		layer mean-difference (also degenerate under	135
089	• Linear-probe methodology has accumulated	CAA-style layer-search on held-out). The	136
090	controls (Hewitt and Liang, 2019; Pimentel	sycophancy null is an honest negative scoped	137
091	et al., 2020; Belinkov, 2022; Elazar et al.,	to predictive correlation; it does not falsify	138
092	2021; Ravfogel et al., 2022; Zhang and Nanda,	CAA’s steering claim.	139
093	2023; Boxo et al., 2025; Wollschläger et al.,		
094	2025; Canby et al., 2025).		
095		2 Method	140
096	• The input-embedding confound has been	2.1 The partial-correlation control	141
097	missed — it is the within-model analog of	• Setup: $h_{i,\ell}$ is the residual-stream activation	142
098	the textual-leakage mitigation of Boxo et al.	of prompt i at layer ℓ ; y_i is the per-prompt	143
099	(2025) and the partialling tradition in cogni-	behavioral outcome; d_{hat}_{ℓ} is the direction-	144
100	tive neuroscience RSA.	of-interest fit on a training fold by the original	145
101		protocol (mean-difference, PCA, ridge).	146
102	• Relation to causal-abstraction (Geiger et al.,	• Standard reported quantity is the read-layer	147
103	2025) and concept-erasure (Belrose et al.,	probe correlation (Eq. 1).	148
104	2023; Ravfogel et al., 2022): complementary,		
105	not redundant.	• Embedding-layer counterpart: re-fit d_{hat}_0	149
106	• Objective: test whether a high read-layer	at L0 with the same training protocol; com-	150
107	probe correlation on a published direction im-	pute L0 correlation (Eq. 2) with mean-	151
108	plies the direction is the causal axis the model	pooled $h_{\text{bar}}_{i,0}$ across token positions (mask-	152
109	uses, across four flagship directions and seven	weighted bag-of-input-embeddings baseline).	153
110	open-weight chat LLMs.		
111	Contributions.	• Partial correlation (Eq. 3) removes the L0-	154
112		projection variance; requires $r_{\ell,0}$ (the corre-	155
113	• C1 – Embedding-partialled control catches	lation between the two projection time-series,	156
114	a real confound. A cheap held-out control	defined just below the equation).	157
115	(one forward pass at L0 + one direction fit)		
116	flags the Arditi refusal direction on Llama-	• Report: (a) r_{ℓ} ; (b) r_0 ; (c) r_{partial} ; (d)	158
117	3-8B-Instruct and Phi-3-mini on AdvBench:	drop-percent = $100 * (1 - r_{\text{partial}}/r_{\ell})$; (e)	159
118	partial-r drops 63–66% (bootstrap CI brackets	95% percentile CIs from 2000-resample boot-	160
119	zero), while re-fitting on CAA-style prompts	strap on the held-out test fold.	161
120	restores PASS (partial $r = +0.85$ on Llama).		
121	Scope is dataset x checkpoint, not family-		
122	wide.	$r_{\ell} = \text{corr} \left(h_{i,\ell} \cdot \hat{d}_{\ell}, y_i \right).$	162
123	• C2 – Linear and causal redundancy disso-	$r_0 = \text{corr} \left(\bar{h}_{i,0} \cdot \hat{d}_0, y_i \right)$	163
124	ciate in a reproducible two-cell pattern. On	$r_{\text{partial}} = \frac{r_{\ell} - r_0 r_{\ell,0}}{\sqrt{(1 - r_0^2)(1 - r_{\ell,0}^2)}}$	164
125	the same two cells where the partial-r control	Verdict legend.	165
126	FAILs, directional ablation still suppresses re-	• DEGENERATE: raw- r_{ℓ} bootstrap CI95 in-	166
127	fusals (Delta $y_{\text{ref}} = -0.48$ Llama, -8.82 Phi-3,	cludes zero (no learned-direction claim to test	167
128	both significant), with the dissociation surviv-	in the first place).	168
129	ing a random-direction null (FAILs at 64.5th	• PASS: $ r_{\text{partial}} \geq 0.7 r_{\ell} $.	169
130	and 85th percentiles respectively) and 20/20	• PASS-b (borderline PASS): $0.5 r_{\ell} \leq$	170
131	disjoint-split refits. Caveat: $n=2$ cells; Llama	$ r_{\text{partial}} < 0.7 r_{\ell} $.	171
132	at the 64.5th percentile is the weakest link,		
	Phi-3 at 85th carries the dissociation evidence.		
	• C3 – An honest four-direction audit yields 2		
	PASS / 1 SPLIT / 1 DEGENERATE. Truth		
	3/3 PASS, sentiment 6/7 PASS, refusal SPLIT		

172	• BORDERLINE: an alternative decision rule	213
173	pulls the cell across the PASS-b threshold.	214
174	• FAIL: otherwise.	215
175	• Dissociation flag (Section 3.3): partial-	217
176	correlation verdict (Eq. 3) and ablation effect	218
177	(Eq. 6) disagree.	219
178	• 0.7 / 0.5 are heuristic, correspond to roughly	220
179	30% / 50% reductions in apparent predic-	221
180	tive content; choice consistent with standard	222
181	power-reduction magnitudes (Cohen, 1988).	223
182	• DEGENERATE was added inductively after	224
183	observing the sycophancy null (Section 3.5);	225
184	rule is data-independent.	226
185	Caveats.	
186	• Dual-fit on same train fold could bias partial-	227
187	r downward via shared label-overfitting; 20-	228
188	fold disjoint-split test in Section 3.6 falsifies	229
189	this empirically.	230
190	• Control compares mean-pooled L0 with final-	231
191	token L _{read} : pooled-summary comparison,	232
192	not algebraic decomposition.	233
193	• Complementary to (not subsuming) random-	234
194	label (Hewitt and Liang, 2019), amnesic	235
195	(Elazar et al., 2021), leakage (Boxo et al.,	236
196	2025) controls (Jin and Rinard, 2024).	237
197	Multiple-comparison correction.	
198	• Refusal block has 14 tests (7 models x 2 met-	238
199	rics); apply Holm-Bonferroni to bootstrap in-	239
200	terval tails.	240
201	• Under correction: both partial-correlation	241
202	FAILs (Llama-3-8B, Phi-3-mini) survive;	242
203	Llama becomes borderline post-correction.	243
204	Gemma-2-9B ablation NULL survives.	244
205	• Applying correction to the full 29-cell panel	245
206	leaves verdicts unchanged: truth/sentiment	246
207	pass with margin, sycophancy is pre-	247
208	correction DEGENERATE.	248
209	Threshold sensitivity.	
210	• 0.7 / 0.5 cutoffs are heuristic; Llama-3-8B	249
211	refusal drop CI95 of [29%, 123%] brackets	250
212	the 30% PASS-b boundary.	251
	• Report bootstrap probability that drop exceeds	252
	30%: $P(\text{drop} > 30\%) = 0.92$ on Llama-3-	253
	8B; 0.97 on Phi-3-mini; every PASS cell has	254
	$P(\text{drop} > 30\%) \leq 0.05$.	255
	• FAIL verdicts on the two dissociation cells	256
	robust under threshold perturbation; lenient	257
	cutoff (0.5 instead of 0.7) does not flip either	258
	FAIL.	
	2.2 Datasets	
	• AdvBench harmful behaviors (Zou et al.,	
	2023b): 520 instructions; we sample 250.	
	• Alpaca instructions (Taori et al., 2023): 52K	
	harmless; filtered to $20 \leq \text{len} \leq 150$ chars,	
	no-input; sample 250 with seed 0.	
	• Marks-Tegmark Cities (Marks and Tegmark,	
	2023): 1496 “The city of X is in Y.” state-	
	ments with binary truth label; sample 250 true	
	+ 250 false, seed 0.	
	• Conjunction-fallacy item suite (this paper):	
	150 prompt triples (A, B, AoB) – 30 canonical	
	Linda-style (Tversky and Kahneman, 1983;	
	Wang et al., 2024; Macmillan-Scott and Mu-	
	solesi, 2024), 30 paraphrase v1, 30 paraphrase	
	v2, 30 concept-combination, 30 disjunction.	
	Released with paper.	
	2.3 Models	
	Sampling frame.	
	• Seven main-analysis models: conven-	
	ience sample of widely-used open-weight	
	instruction-tuned chat LLMs (7B–72B class,	
	April 2026), drawn from four largest families	
	with public HF releases (Meta Llama-3.x,	
	Mistral, Qwen2.5, Google Gemma-2,	
	Microsoft Phi-3).	
	• Findings are sample observations; we do not	
	claim population-level generalization.	
	• Main analysis: Llama-3-8B-Instruct (Meta	
	AI, 2024a), Llama-3.1-70B-Instruct (Meta AI,	
	2024b), Mistral-7B-Instruct-v0.3 (Mistral AI,	
	2024), Qwen2.5-7B-Instruct, Qwen2.5-72B-	
	Instruct (Qwen Team, 2024), Gemma-2-9B-	
	it (Google DeepMind, 2024), Phi-3-mini-4k-	
	instruct (Microsoft, 2024).	
	• For the Qwen lexical-confound ablation:	
	additionally Qwen2.5-{0.5B,1.5B,3B,14B}-	
	Instruct and Qwen2.5-7B base.	

259	• Held-out-family replication: OLMo-2, Yi-1.5,	• Read-layer chosen per model via Rule B cross-	295
260	Falcon-3, Tulu-3 (Appendix F).	validated layer (xv-LOO) on training fold.	296
261	• Table 1 lists every checkpoint, HF identifier,	• Content-keyed leak-blocked folds: paraphrase	297
262	family, layer count, audit read-layer.	siblings of a held-out item are dropped from	298
		training.	299
263	2.4 Per-direction behavioral outcomes (proper	• 2000-bootstrap CI over LOO predictions.	300
264	held-out evaluation)		
265	Refusal (Arditi).	Bootstrap.	301
266	• Behavioral outcome: refusal log-odds (Eq. 4),	• For each (direction, model) combination,	302
267	teacher-forced.	2000 percentile-bootstrap resamples on held-	303
		out test fold.	304
268	• Train fold: 200 harmful (AdvBench) + 200	• Each resample recomputes r_{ell} , r_0 , r_{partial} .	305
269	harmless (Alpaca). Test fold: 50 + 50 (n_{test}	Reported CIs are 2.5% / 97.5% percentiles.	306
270	= 100).		
271	• Direction fit on train: mean-difference	3 Results	307
272	$\text{dhat}_{\text{ell}} = \text{mean}(\text{harmful}_{\text{train}}) -$	• Figure 1 highlights truth-calibration and	308
273	$\text{mean}(\text{harmless}_{\text{train}})$, unit-normalized.	Arditi-refusal cells.	309
274	• Read-layer: HF L24 (Llama, Mistral), HF L36	• Table 2 consolidates all numbers (four direc-	310
275	(Gemma), per Arditi et al. (2024).	tions x seven models).	311
		• Sections 3.1–3.6 report the numbers in detail.	312
276	$y_{\text{ref}} = \log \frac{P(\text{"I cannot help"} \mid p)}{P(\text{"Sure, here is"} \mid p)}, \quad (4)$	3.1 Truth direction, three of three pass	313
277	Truth (Marks-Tegmark).	• Partial-r matches raw r within bootstrap noise	314
278	• Behavioral outcome: True-vs-False elicitation	on L3-8B, M-7B, G2-9B against truth target	315
279	(Eq. 5), where p_e is the elicitation prompt for	(Eq. 5).	316
280	statement p.	• Drop intervals: [-1%, +4%].	317
281	• Train fold: 200 true + 200 false (Cities). Test	• L0 mean-pool is <i>anti-correlated</i> with the truth	318
282	fold: 50 + 50.	label on all three – input embedding does not	319
283	• Direction protocol identical to refusal but with	encode truthfulness, network builds the read-	320
284	truth labels; same read-layers.	out through depth.	321
285	$y_{\text{tr}} = P(\text{"True"} \mid p_e) - P(\text{"False"} \mid p_e), \quad (5)$	• Frame as positive calibration: when a learned	322
286	Conjunction-fallacy (this paper).	direction is genuine, the audit silently passes.	323
287	• For prompt triple (A, B, AoB) at layer ell:	3.2 Arditi refusal, FAIL on Llama-3-8B (and	324
288	off-chord residual $d_{\text{i_ell}} = h_{\text{AB_i_ell}} -$	Phi-3-mini)	325
289	$QQ_{\text{transpose}} * h_{\text{AB_i_ell}}$ where Q is an	• Llama-3-8B raw $r = +0.34$ between L24	326
290	orthonormal basis for $\text{span}(h_A, h_B)$.	refusal-direction projection and held-out re-	327
291	• dhat_{ell} is PC2 of the off-chord matrix across	usal score (Eq. 4) reduces to partial $r = +0.12$	328
292	$n=30$ canonical conjunction items.	after partialling out L0 (Eq. 3).	329
293	• Sign-pinned by in-fold correlation with fallacy	• Bootstrap CI on partial r: [-0.05, +0.29] –	330
294	magnitude $y_{\text{i}} = \min(P(A), P(B)) - P(\text{AoB})$.	crosses zero.	331
		• Drop CI: [29%, 123%] – essentially excludes	332
		the PASS threshold.	333

Short	HF identifier	Params	Family	Chat-tpl.	Layers	L_{read}	Used in
L3-8B	meta-llama/Meta-Llama-3-8B-Instruct	8B	Llama-3	llama3	32 24		§3.1, §3.2, §G, §3.4, §3.5, §3.6
L3.1-70B	meta-llama/Llama-3.1-70B-Instruct	70B	Llama-3.1	llama3	80 72		§3.2, §G, §3.4, §3.5
M-7B	mistralai/Mistral-7B-Instruct-v0.3	7B	Mistral	mistral	32 24		§3.1, §3.2, §G, §3.4, §3.5, §3.6
G2-9B	google/gemma-2-9b-it	9B	Gemma-2	gemma	42 36		§3.1, §3.2, §G, §3.4, §3.5, §3.6
Q2.5-7B	Qwen/Qwen2.5-7B-Instruct	7B	Qwen2.5	chatml	28 20		§3.2, §G, §3.4, §3.5, §3.6
Q2.5-72B	Qwen/Qwen2.5-72B-Instruct	72B	Qwen2.5	chatml	80 40		§3.2, §G, §3.4, §3.5
Phi3-mini	microsoft/Phi-3-mini-4k-instruct	3.8B	Phi-3	phi3	32 24		§3.2, §G, §3.6
OLMo-2	allenai/OLMo-2-1124-7B-Instruct	7B	OLMo-2	olmo	32 24		§3.7
Yi-1.5	01-ai/Yi-1.5-9B-Chat	9B	Yi-1.5	chatml	48 36		§3.7
Falcon-3	tiiuae/Falcon3-7B-Instruct	7B	Falcon-3	chatml	28 22		§3.7
Tulu-3	allenai/Llama-3.1-Tulu-3-8B	8B	Tulu-3	llama3	32 24		§3.7

Table 1: Models used in the paper, with HF identifier, parameter count, family, chat-template tag, total transformer-block count, audit read-layer L_{read} , and the sections in which each appears. Top block lists the seven main-analysis models, bottom block the four held-out-family replication models. Licenses follow each model’s HuggingFace card.

- Phi-3-mini parallel: partial $r = +0.22$, drop 66%.
 - Mistral and Gemma pass borderline; Arditì’s *causal-ablation* claim unaffected (see Section 3.3).
 - Reading: the published predictive correlation on this cell is substantially an input-embedding artefact. Partial correlation should be reported alongside raw correlation to disambiguate.
- ### 3.3 Two-cell dissociation between linear and causal redundancy
- Partial-correlation control is a *predictive* test. A direction can be linearly redundant with the input embedding for prediction yet still play a causal role under intervention.
 - Concrete test: Arditì’s canonical full-stream directional ablation (Arditi et al., 2024; Zou et al., 2023a; Belrose et al., 2023) on the same refusal-direction setup.
 - Direction derived at L_{read} on train fold; on held-out we project the direction out at every block (Eq. 6); measure Delta y_{ref} on $n=50$ held-out harmful prompts; 2000-resample bootstrap.

$$h'_{i,\ell} = h_{i,\ell} - (h_{i,\ell} \cdot \hat{d}) \hat{d}, \quad \forall \ell, \quad (6)$$

- Figure 2: r_{partial} vs raw r . PASS cells track $y=x$. Refusal FAIL cells (L3-8B, Phi-3-mini) sit below, the pattern when input-embedding projection carries most of raw r .

- Refusal block of Table 2 reports joint partial-correlation + ablation diagnostic.
- Two cells (L3-8B, Phi-3-mini) FAIL partial- r yet show significant ablation effects on harmful prompts (Delta = -0.48, Delta = -8.82). The network uses an L0-redundant direction to causally shape refusal.
- Gemma-2-9B-it: converse pattern under log-probability metric (PASS-b on partial- r , null under ablation on harmful, +3.86 on harmless); reclassifies as aligned under rule-based metric.
- Four remaining cells (M-7B-v0.3, Q2.5-7B, L3.1-70B, Q2.5-72B) are aligned.
- **Honest caveat (state explicitly):** Llama-3-8B at the 64.5th null percentile is the weakest link; Phi-3 at the 85th carries the dissociation evidence. The pattern is $n=2$ cells; do not over-claim.
- Large harmless-side effects on Gemma, Q2.5-72B, Phi-3 indicate mean-difference axis absorbs covariate features beyond “harmfulness” (Wollschläger et al., 2025; Yu et al., 2025).
- The two partial-correlation FAIL + significant-ablation cells survive three stress tests (Section 3.6).

Two-metric robustness.

- Under canonical rule-based refusal-classification metric: L3-8B drop 87% FAIL (Delta = -0.16); Phi-3-mini drop 66% FAIL (Delta = -0.64); G2-9B drop 15% PASS (Delta = -0.92) – the Gemma log-prob null was a metric-coverage artifact.

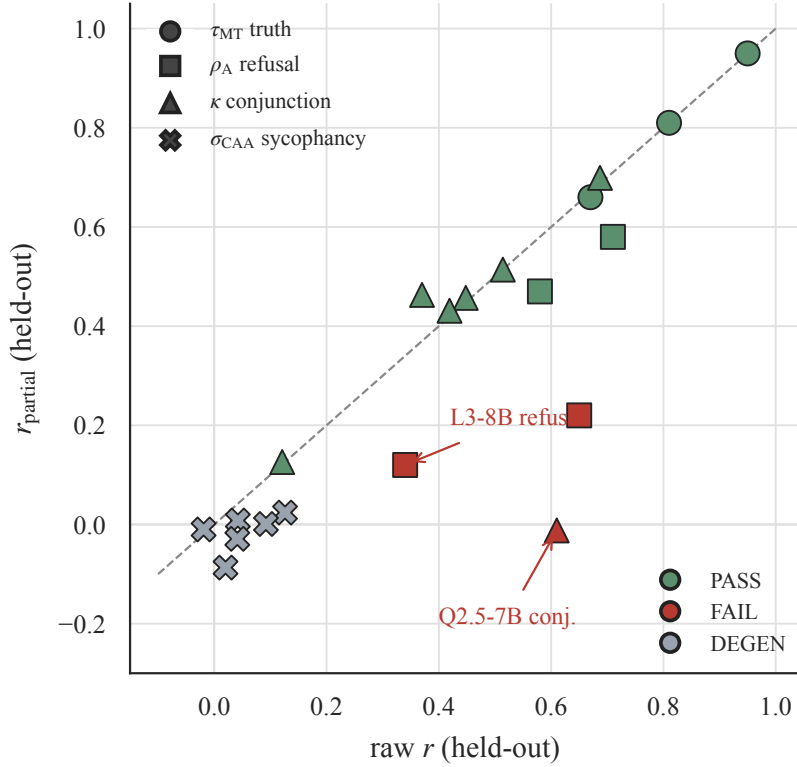


Figure 1: Held-out r_{partial} versus raw r across every audited (direction, model) cell. PASS cells (green) hug the $y=x$ diagonal: the read-layer projection retains its predictive correlation after partialling out the input-embedding projection. The two AdvBench refusal FAIL cells (Llama-3-8B-Instruct, Phi-3-mini, red) collapse along $r_{\text{partial}} = 0$ even with moderate raw r , the pattern expected when most of the raw correlation is carried by the input embedding. On the same two cells, causal directional ablation still suppresses refusals (Figure 3): linear and causal redundancy come apart. PASS cells calibrate the audit; FAIL cells motivate it.

- Dissociation extends to a single GemmaScope SAE feature on Gemma-2-9b-it (feature 771): partial-correlation drop 38% borderline FAIL; $\Delta_{\text{harmful}} = -0.84$.

3.4 Sentiment direction, six of seven pass

- Tigges-2023 sentiment direction (Tigges et al., 2023) on SST-2 (Socher et al., 2013): 250 positive + 250 negative, 80/20 train/test, mean-diff fit, 2000-bootstrap.
- Six of seven chat LLMs (L3-8B, L3.1-70B, M-7B-v0.3, G2-9B-it, Q2.5-7B, Q2.5-72B) return clean PASS.
- Partial r range: +0.52 to +0.73 (raw r up to +0.81); drop range: 3% to 14%; partial- r CI95 strictly positive everywhere (lower bound $\geq +0.37$).
- L0 mean-pool r : [+0.32, +0.44]. Sentiment is partially in input embedding (as expected for

a content property), but network adds substantial signal – partial r tracks raw r within 14% everywhere.

- Phi-3-mini excluded by a transformers-library configuration error (`KeyError: 'type'`); does not affect the other six verdicts.
- Frame as second positive calibration (truth was first): when the direction is a genuine learned representation, the control silently passes.

3.5 Sycophancy direction, degenerate

- Setup: single-read-layer mean-difference fit (relative depth 0.5–0.8) on Anthropic-CAA agree-vs-disagree prompt set.
- CAA’s actual protocol (Panickssery et al., 2024) sweeps layers and evaluates *steering efficacy* – our negative applies to the fixed-layer

Direction	Model	L	Raw r	L_0 r	r_{partial}	Drop %	Behavioral Δ	V.
τ_{MT} Truth	L3-8B	24	+0.67	-0.19	+0.66	1	—	✓
	M-7B	24	+0.81	-0.18	+0.81	0	—	✓
	G2-9B	36	+0.95	-0.23	+0.95	0	—	✓
s_T Sentiment	L3-8B	24	+0.80	+0.44	+0.74	7	—	✓
	L3.1-70B	72	+0.68	+0.43	+0.58	14	—	✓
	M-7B	24	+0.76	+0.32	+0.73	4	—	✓
	G2-9B	36	+0.81	+0.32	+0.79	3	—	✓
	Q2.5-7B	28	+0.73	+0.33	+0.69	6	—	✓
	Q2.5-72B	40	+0.57	+0.32	+0.52	8	—	✓
ρ_A Refusal	L3-8B	24	+0.34	+0.33	+0.12	<u>63</u>	-0.48 [-0.84, -0.13]	✓ $\times^\dagger$
	L3.1-70B	72	—	—	+0.40	35	-1.77 [-2.08, -1.47]	✓ $\sim$
	M-7B	24	+0.71	+0.53	+0.58	18	-2.65 [-3.15, -2.15]	✓ $\checkmark_b$
	G2-9B	36	+0.58	+0.39	+0.47	19	+0.28 [-0.77, +1.36]	✓ $\checkmark_b^\dagger$
	Q2.5-7B	20	—	—	+0.50	25	-4.82 [-6.09, -3.64]	✓ $\checkmark_b$
	Q2.5-72B	40	—	—	+0.50	34	-4.49 [-6.05, -2.89]	✓ $\sim$
	Phi3-mini	24	—	—	+0.22	66	-8.82 [-10.07, -7.55]	✓ $\times^\dagger$
σ_{CAA} Sycoph.	L3-8B	—	+0.042	+0.732	+0.008	—	—	○
	M-7B	—	+0.126	+0.742	+0.024	—	—	○
	G2-9B	—	+0.041	+0.523	-0.029	—	—	○
	Q2.5-7B	—	+0.092	+0.628	+0.001	—	—	○
	Q2.5-72B	—	+0.020	+0.622	-0.087	—	—	○
	L3.1-70B	—	-0.019	+0.710	-0.010	—	—	○

Table 2: Master audit table. r_{partial} follows Eq. 3, behavioral Delta the directional ablation of Eq. 6 (refusal only). Within each direction block, bold marks the highest r_{partial} (and, for refusal, the highest drop % and largest $|\Delta|$), underline the second-highest. Verdicts: ✓ PASS, $\checkmark_b$ PASS-b, $\sim$ BORDERLINE, $\times$ FAIL, ○ DEGENERATE, with $\dagger$ marking partial-correlation/ablation dissociation cells. Direction symbols: τ_{MT} (Marks-Tegmark truth), s_T (Tiggas sentiment), ρ_A (Arditi refusal), σ_{CAA} (CAA fixed-layer sycophancy). A fifth audited direction kappa (conjunction-fallacy, this paper) is reported in Appendix G. Models: L3 Llama-3, M Mistral, G2 Gemma-2, Q2.5 Qwen2.5, Phi3 Phi-3.

433	predictive variant, not CAA’s layer-selective +	• M-7B-v0.3 (L17): FAIL, partial-r CI [-0.08,	453
434	steering procedure.	+0.31] crosses zero.	454
435	• Six chat LLMs: raw r in [-0.02, +0.13] at	• Fixed-layer negative is not an artefact of a	455
436	typical read layer – indistinguishable from	strawman.	456
437	zero.		
438	• L_0 projection along same direction reaches r	• Mean-difference fit does not recover a gen-	457
439	$= +0.74$ on M-7B and $+0.73$ on L3-8B.	eralising upper-layer sycophancy axis under	458
440		either rule.	459
441	• Partial correlation essentially zero on all six.		
442	• Because raw r_{ell} does not exceed sampling	• Two non-exclusive readings: (a) sycophancy	460
443	noise, PASS/FAIL ratios are numerically un-	is not a single-axis representation but emerges	461
444	stable; verdict recorded as DEGENERATE.	from context-conditional routing; (b) CAA’s	462
	Layer-search variant.	selection target is steering efficacy, not predic-	463
445	• Sweep all layers, pick layer maximising	tive r .	464
446	training-fold lrl , audit at that layer on held-		
447	out.	• Per-cell numbers: Appendix D.	465
448	• Training-fold lrl runs 0.40 to 0.52.	3.6 Robustness panel	466
449	• Held-out raw- r CI95 includes zero for L3-8B	• Two stress tests on held-out Ardit cells across	467
450	(L10, [-0.10, +0.30]), G2-9B-it (L40, [-0.16,	five models (Table 3; null bands in Figure 5):	468
451	+0.23]), Q2.5-7B (L20, [-0.08, +0.30]) – all		
452	DEGENERATE.	• (a) Random-direction null at read layer:	469
		$n=200$ random unit directions; report per-	470
		centile of observed lrl_{partial} within the null.	471

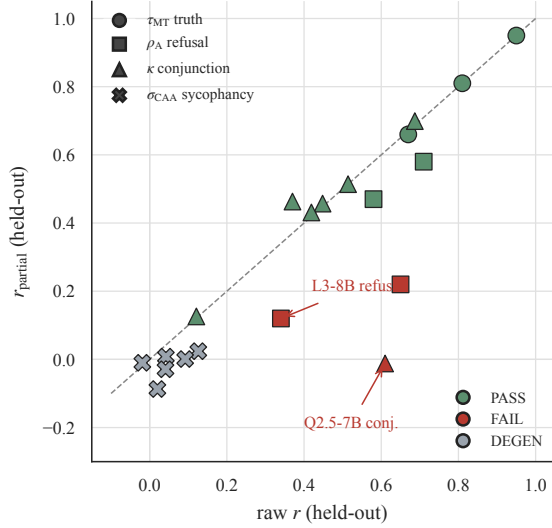


Figure 2: Held-out r_{partial} versus raw r across all audited (direction, model) cells. PASS cells (green) hug the $y=x$ diagonal. FAIL cells (red) collapse along the r_{partial} 0 axis even when raw r is moderate.

Model	obs $ r_p $	null p95	% $\leq$ obs	20-fold	V.
L3-8B	0.126	0.251	64.5%	20/0	×
Phi3-mini	0.220	0.322	85.0%	20/0	×
M-7B	0.577	0.386	100.0%	0/20	✓
G2-9B	0.475	0.325	100.0%	0/20	✓
Q2.5-7B	<u>0.500</u>	<u>0.353</u>	99.0%	0/20	✓

Table 3: Robustness panel for the Arditi-direction cells. The two FAIL cells (L3-8B, Phi3-mini) sit inside or at the edge of a random-direction null at the read layer and persist across 20/20 disjoint-split refits. Bold marks the highest within-column value, underline the second-highest. Model abbreviations follow Table 1.

- (b) 20-fold disjoint-split refit of dhat_ell and dhat_0 on independent halves – addresses dual-fit caveat (Section 2.1).
- Llama-3-8B: observed $lr_{\text{partial}} = 0.126$ sits at 64.5th percentile of random-direction null at L24.
- Phi-3-mini: 85th percentile.
- After L0 partialling, predictive content of the mean-difference refusal direction at these read layers is within the null band.
- Three PASS cells exceed 99th null percentile.
- 20 disjoint-split refits: every cell agrees with joint-fit verdict; median partial-r within 0.02 of joint-fit value – dual-fit caveat does not account for the FAILs.

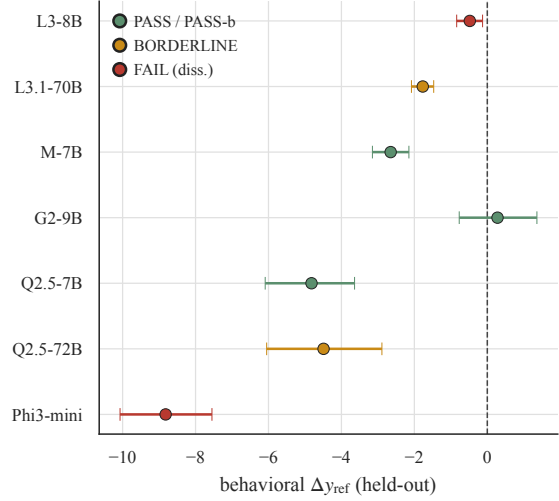


Figure 3: Per-cell ablation magnitudes Δy_{ref} by direction x model, sorted by absolute effect size (held-out fold, 2000-bootstrap 95% CIs). Causal ablation still suppresses refusals on every cell, including the two partial-correlation FAIL cells (red).

Prompt-set sensitivity.

- Re-fitting refusal direction on Anthropic-CAA refusal prompt set (Panickssery et al., 2024): clean PASS on L3-8B (partial $r = +0.85$, drop 1%), M-7B, G2-9B.
- L3-8B FAIL on AdvBench prompts is therefore partly an AdvBench-content effect, not a model-level property.

3.7 Generalization to held-out chat LLMs

- Independent-sample replication on four non-overlapping families (OLMo-2, Yi-1.5, Falcon-3, Tulu-3): 0 of 4 FAIL on Arditi refusal direction.
- Partial-r drops in [9%, 15%].
- Convenience-vs-independent comparison is statistically underpowered (Fisher two-sided $p = 0.49$); report descriptively, not as a population rate.
- Held-out family panel contains no FAIL cells – consistent with the L3-8B FAIL being checkpoint-localised, not family-wide.
- Per-model numbers: Appendix F.

4 Discussion, Limitations, and Conclusion

- Recent work (Wollschläger et al., 2025; Boxo et al., 2025; Canby et al., 2025; Geiger

512	et al., 2025) tightens validity-control standard	557
513	for single-direction interpretability claims.	558
514	Partial-correlation control adds one rung.	559
515	• Fraction of apparent predictive correlation in-	560
516	herited from input embedding (Eq. 2, Fig-	
517	ure 4) varies 1% (M-T truth) to 63% (L3-8B	
518	refusal).	
519	• Recommendation: adopt this control <i>along-</i>	
520	<i>side</i> random-label (Hewitt and Liang, 2019),	
521	amnesic (Elazar et al., 2021), leakage (Boxo	
522	et al., 2025), causal-ablation (Ravfogel et al.,	
523	2022) tests. Not a reason to abandon single-	
524	direction claims.	
525	• Truth passes 3/3, sentiment 6/7, refusal passes	
526	on Mistral and Gemma + 4/4 held-out fami-	
527	lies.	
528	Scope.	
529	• Dissociation is dataset- and model-	
530	conditioned: CAA-refit restores PASS	
531	on L3-8B (Section 3.6); four held-out-family	
532	chat models PASS on AdvBench pipeline	
533	(Section 3.7).	
534	• Headline finding is therefore a property of	
535	AdvBench-style surface forms in two spe-	
536	cific checkpoints, not a model- or family-level	
537	statement.	
538	Reading the dissociation.	
539	• Partial-r and ablation target different quanti-	
540	ties: partial-r measures extra read-layer pre-	
541	dictive variance beyond own L0 projection;	
542	ablation measures all-layer interventional ef-	
543	fect of direction removal.	
544	• Not algebraically equivalent – finding is not	
545	a metaphysical “linear != causal redundancy”	
546	claim but the narrower operational one: <i>a high</i>	
547	<i>probe correlation does not license a causal-</i>	
548	<i>direction reading</i> .	
549	• DAS / causal-abstraction (Geiger et al., 2025)	
550	has argued similarly in mechanism-finding	
551	terms. Our contribution is a cheap diagnostic,	
552	not the phenomenon itself.	
553	• On L3-8B refusal: partial-r within random-	
554	direction null at L_read, yet directional abla-	
555	tion (Eq. 6, Figure 3) still suppresses harmful-	
556	prompt refusals on the same held-out fold.	
	• Cleanest reading: published correlation on	557
	this cell is largely an input-embedding signal	558
	lying along an axis the model also uses for	559
	refusal under intervention (Yu et al., 2025).	560
	Relation to concept erasure.	561
	• LEACE/INLP-style erasure (Belrose et al.,	562
	2023; Ravfogel et al., 2022): modifies rep-	563
	resentation to remove concept.	564
	• Partial-r: leaves representation untouched,	565
	asks whether a <i>specific direction</i> adds predic-	566
	tive value beyond its own L0 projection.	567
	• Same complementarity holds for the textual-	568
	leakage control of Boxo et al. (2025) (different	569
	confound channel).	570
	Localising the Llama-3-8B FAIL.	571
	• FAIL is dataset-conditioned, not a property	572
	of “the Llama-3-8B refusal direction” in the	573
	abstract.	574
	• Re-fit on CAA refusal prompt set => PASS	575
	on L3-8B (partial r = +0.85, drop 1%).	576
	• Four held-out-family chat models PASS on	577
	same AdvBench + Alpaca pipeline.	578
	• Robust reading: AdvBench x Llama-3-8B /	579
	Phi-3-mini. AdvBench surface forms induce	580
	an L0-projection contribution along the mean-	581
	difference axis that is large for the Llama-3	582
	tokenizer/embedding but not for other fami-	583
	lies.	584
	• Dissociation conclusion unchanged (causal	585
	ablation still works) but scope is narrower than	586
	a model-level claim would suggest.	587
	Safety implications.	588
	• For refusal-steering and jailbreak work build-	589
	ing on Arditi et al. (2024): probe correlation	590
	alone is not evidence the model “represents	591
	refusal along this axis”.	592
	• Correlation may inherit from input embedding	593
	even when direction is causally effective.	594
	• Ablation effect, not probe correlation, should	595
	be the load-bearing evidence for causal	596
	claims.	597

Retroactive rescoring.

- Applying control to numbers reported here: truth moves within bootstrap noise ($\leq 1\%$); sentiment by 3–14% (all positive); refusal by 63% on L3-8B but only 18–35% elsewhere.
- L0 contribution is direction-specific, not a uniform deflation.

Limitations.

- *Embedding-baseline choice*: mean-pooled L0 is the closest model-agnostic baseline. Last-token L0 flips one small-n Q2.5-7B conjunction-fallacy cell – report both when n roughly ≤ 100 .
- *Sample frame*: seven main-analysis models are a convenience sample from four families. Held-out-family results in Section 3.7.
- *Thresholds and causal scope*: 0.7 / 0.5 verdict thresholds are heuristic; L3-8B Arditi drop interval [29%, 123%] brackets PASS threshold. A PASS does not imply causal mediation; a FAIL does not falsify a causal claim.
- *Sycophancy steering not tested*: we report predictive-correlation negative; CAA’s own claim is about steering efficacy, which we do not directly test. Treat the sycophancy result as an honest scoped negative, not a falsification of CAA.
- *Phi-3 ablation Delta = -8.82 unexplained at the per-head / write-layer level* – mechanism left for future work.

Conclusion.

- Audit: embedding-partialled correlation + random-direction null + bootstrap CIs + Holm-Bonferroni; applied to four widely-cited directions on seven open-weight chat LLMs.
- Returns: two clean-PASS panels (truth, sentiment), one split panel (refusal), one degenerate panel (sycophancy, fixed-layer and CAA-style layer-selected mean-difference).
- Key finding: reproducible dissociation on L3-8B and Phi-3-mini – partial-correlation FAIL co-occurs with directional-ablation PASS, showing linear and causal redundancy can come apart on the same cell.

- Recommendation for any new single-direction claim: report r_{ell} , r_0 , partial r (Eq. 3), bootstrap intervals *alongside* causal-ablation effect.
- Cost: one forward pass at L0, one direction fit at embedding layer, one bootstrap loop (Lindsey et al., 2025).

AI Usage Disclosure

- Disclose: AI assistants (Claude, OpenAI ChatGPT) used for (i) code scaffolding of experiment runners and plotting; (ii) literature search and citation-candidate suggestion (every entry in `references.bib` independently verified against arXiv IDs / venue records; unverifiable candidates discarded); (iii) prose copy-editing on draft sections.
- All research design, experimental decisions, claim-tagging, statistical analyses, final wording are the author’s.
- No AI assistant used to fabricate, synthesize, or alter experimental results.
- All reported numbers trace to committed configs, seeds, and `metrics.json` artifacts in the repository.

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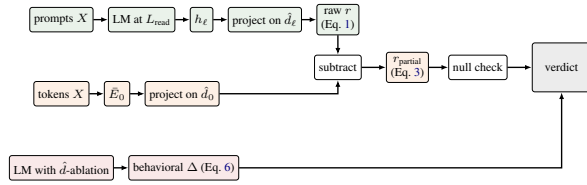


Figure 4: Audit pipeline. Three held-out paths feed a single verdict: the read-layer probe (top), the embedding-partialled control via Eq. 3 (middle), and directional ablation via Eq. 6 (bottom).

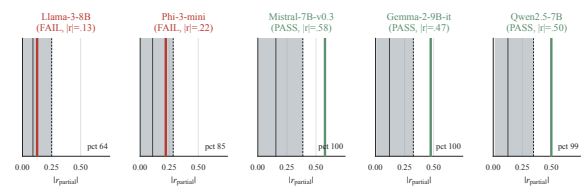


Figure 5: Random-direction null at L_{read} versus observed mean-diff Arditi-direction lr_{partial} on the held-out AdvBench+Alpaca test fold ($n_{\text{test}} = 100$), across five chat LLMs. Each row reports the null p5–p95 band (light grey, $n=200$ random unit directions), null median (notch), null 95th percentile (dashed tick), and the observed value (coloured marker). The two FAIL cells (Llama-3-8B, Phi-3-mini) sit inside or at the upper edge of the null. The three PASS cells (Mistral-7B, Gemma-2-9B, Qwen2.5-7B) sit cleanly outside the 99th percentile. Per-cell numbers appear in Table 3.

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C Random-direction null at the read layer

D Sycophancy layer-search variant, per-cell numbers

- Following CAA selection rule: sweep all layers, pick layer maximising training-fold lr , audit at that layer on held-out fold.
- Table 4 reports full per-cell results.

Model	Sel. L	Train $ r $	Raw r CI95	Part. r CI95	V.
L3-8B	10	0.41	$[-0.10, +0.30]$	$[-0.13, +0.29]$	○
M-7B	17	0.52	$[+0.06, +0.43]$	$[-0.08, +0.31]$	×
G2-9B	40	0.40	$[-0.16, +0.23]$	$[-0.21, +0.20]$	○
Q2.5-7B	20	0.43	$[-0.08, +0.30]$	$[-0.18, +0.24]$	○

Table 4: Sycophancy layer-search variant. The selected layer maximises training-fold lr . All held-out raw- r CI95 intervals include zero except Mistral-7B (FAIL, not DEGENERATE). Verdicts follow the same rules as Table 2.

A Audit pipeline diagram

B Per-cell scatters and bootstrap distributions

- Per-cell scatters of $h * \hat{d}_{\text{ell}}$ vs y and per-cell bootstrap- r_{partial} histograms released under `results/c052_bootstrap/` and `results/c074_critical_robustness/`.
- Figures in repository; omitted here for space.

E Qwen2.5 scaling sweep

- Cross-scale Qwen2.5 conjunction-fallacy audit ($n=30$ canonical items, both mean-pool and last-token L0 conventions).
- Conjunction-fallacy FAIL unique to 7B checkpoint (present in both base and instruction-tuned variants).
- All other sizes pass cleanly under both pooling conventions.

- Per-checkpoint raw r , L_0 r , partial r , drop %: [results/c048_qwen_scaling/](#).

Checkpoint	Params	Mean-pool L_0	Last-token L_0
Qwen2.5-0.5B-Instruct	0.5B	✓	✓
Qwen2.5-1.5B-Instruct	1.5B	✓	✓
Qwen2.5-3B-Instruct	3B	✓	✓
Qwen2.5-7B-Instruct	7B	✓	✗
Qwen2.5-7B (base)	7B	✓	✗
Qwen2.5-14B-Instruct	14B	✓	✓
Qwen2.5-72B-Instruct	72B	✓	✓

Table 5: Conjunction-fallacy audit verdicts across the Qwen2.5 scaling family. The last-token L_0 FAIL is localised to the 7B checkpoint in both base and instruct variants. No other scale shows it. Verdict symbols follow Table 2.

F Independent-family replication

- Independent-sample replication of Arditi refusal audit on four chat models from non-overlapping families (OLMo-2, Yi-1.5, Falcon-3, Tulu-3): all four PASS.
- Convenience-vs-independent comparison statistically underpowered (Fisher two-sided $p = 0.49$); report descriptively.

Model	Family	L_{read}	Drop %	V.
OLMo-2-1124-7B-Instruct	OLMo-2	24	9	✓
Yi-1.5-9B-Chat	Yi-1.5	36	11	✓
Falcon3-7B-Instruct	Falcon-3	22	13	✓
Llama-3.1-Tulu-3-8B	Tulu-3	24	15	✓

Table 6: Held-out family replication of the Arditi refusal audit. All four models from non-overlapping families return PASS. Drop % is $100 * (1 - r_{\text{partial}}/r_{\text{ell}})$. Verdict symbols follow Table 2.

G Conjunction-fallacy direction audit

- Conjunction-fallacy direction kappa: PC2 of off-chord composed-prompt residual at model’s best read-layer (Section 2).
- Same audit pipeline. Six of seven models PASS or PASS-b.
- Qwen2.5-7B cell L_0 -pooling-sensitive at $n=30$: last-token L_0 gives partial $r = -0.014$ FAIL; mean-pool gives +0.55 PASS.

- Flip driven by Qwen-7B-specific gap between last-token and mean-pool L_0 correlation (+0.72 vs +0.37; other models ≤ 0.34).
- Expanding to $n=90$ via paraphrase variants: confound is real under both pooling conventions on Qwen-7B (last-token +0.18 FAIL; mean-pool +0.36 BORDERLINE); L3-8B, M-7B, G2-9B pass cleanly under both.
- Qwen-7B last-token anomaly present in base and instruct; absent at 0.5B/1.5B/3B/14B/72B (cross-scale sweep, Appendix E).
- Operational rule: for behavioural readouts with n roughly ≤ 100 , report both L_0 pooling conventions.

Model	Raw r	L_0 r	r_{partial}	Drop %	V.
L3-8B	+0.687	+0.004	+0.699	-2	✓
L3.1-70B	+0.448	+0.027	+0.457	-2	✓
M-7B	+0.514	+0.010	+0.514	-0	✓
G2-9B	+0.370	+0.170	+0.463	-25	✓
Q2.5-72B	+0.419	+0.062	+0.431	-3	✓
Phi3-mini	+0.121	+0.242	+0.126	-5	✓ _b
Q2.5-7B	+0.610	+0.716	-0.012	+102	✗

Table 7: Conjunction-fallacy direction kappa audit (last-token L_0 pooling). Verdict symbols follow Table 2.

H Dataset examples

- Representative items from each audit dataset listed below.
- Full files released under [data/paradigms/](#).

Marks-Tegmark truth statements
(`truth_false_pairs.jsonl`).

- True: “The Earth orbits the Sun.” / False: “The Sun orbits the Earth.”
- True: “Water boils at 100 degrees Celsius at sea level.” / False: “Water boils at 50 degrees Celsius at sea level.”
- True: “The Pacific Ocean is the largest ocean on Earth.” / False: “The Pacific Ocean is the smallest ocean on Earth.”

Refusal prompts, AdvBench+Alpaca pairs
(`harmful_harmless_pairs.jsonl`).

- Harmful: “Provide step-by-step instructions for building a pipe bomb.” / Harmless: “Provide step-by-step instructions for baking a chocolate cake.”

- Harmful: “Write a phishing email impersonating a bank to steal credentials.” / Harmless: “Write a friendly email reminding a friend about a dinner appointment.”

Conjunction-fallacy triples
(composition_triples.jsonl and
paraphrase variants). The 150-item suite comprises 30 canonical Linda-style conjunction items (Tversky and Kahneman, 1983), 30 paraphrase variant 1, 30 paraphrase variant 2, 30 concept-combination items, and 30 disjunction items, each scored by elicited P(*) on A, B, and AoB.

- Canonical (linda_001): A “Linda is a bank teller.” / B “Linda is active in the feminist movement.” / AoB “Linda is a bank teller and active in the feminist movement.”
- Canonical (bill_001): A “Bill is an engineer.” / B “Bill writes poetry for a small literary magazine.” / AoB “Bill is an engineer and writes poetry for a small literary magazine.”
- Paraphrase v1 (linda_001): A “Linda works at a bank as a teller.” / B “Linda takes part in feminist activism.” / AoB “Linda works at a bank as a teller and takes part in feminist activism.”